



Capstone Project 1
CMU-SE 450

Project Plan
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Online Clinic Booking System

Submitted by

**Truong Gia Huy
Ngo Van Kha
Le Van Xuan
Tran Nguyen Thanh Hong**

**Capstone Project 1 – Mentor:
Master Nguyen Tan Thuan
Signature**

Date:

A handwritten signature in black ink, appearing to read 'Thuan', written over a horizontal line.

Project Information

Project acronym	OnClinic		
Project Title	Online Clinic Booking System		
Start Date	23 Aug 2021	End Date	12 Dec 2021
Lead Institution	International School, Duy Tan University		
Project Mentor	Master Thuan, Nguyen Tan Email: thuannt.it.dtu@gmail.com		
Scrum master / Project Leader & contact details	Ngo Van Kha – 24211210573 Email: ngokha437@gmail.com Tel: 0935950384		
Partner Organization	N/A		
Team Members			
Student ID	Name	Email	Tel
24211206450	Truong Gia Huy	truonggiahuy7799@gmail.com	0972173687
24211207159	Le Van Xuan	levanxuan5518@gmail.com	0905596460
24207106137	Tran Nguyen Thanh Hong	thanhhong24102000@gmail.com	0782363961

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Team members	Name:
	Truong Gia Huy
	Le Van Xuan
	Ngo Van Kha
	Tran Nguyen Thanh Hong
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1. PROJECT OVERVIEW

1.1. Project Description

Project code	OnClinic-C1SE.37	Contract type	Internal Project
Customer	Clinic/Hospital, Patient	End-User	Patient, Doctor
Project Type	Internal	Project Manager/ Scrum master	Ngo Van Kha
Project Category	Lifecycle	Business domain	Health Care
Application type	Application Android		

1.2. Scope and Purpose

Online Clinic Booking System - OnClinic - means an application to book appointments at clinics and examine patients online. It uses intelligent AI technology, which helps to measure human health indicators. From there will give the state of human health. In addition, the application will help users book an appointment to avoid waiting for the clinic.

We use Agile Scrum to develop the project and spend 17 weeks

In this project scope, we just implement these features about hospital, clinic, patient, patient's relatives.

Book an appointment for medical examination offline

- Users enter the application to find a clinic
- Based on the reviews of the clinics, to choose the clinic you want to book an appointment with
- Make an appointment for a clinic
- Go to the clinic/hospital and get medical examination

Book a medical appointment online

- Choose online examination

- Choose the clinic/hospital you want to visit
- Conduct an online examination

Language:

- Vietnamese

1.3. *Assumptions and Constraints*

No	Description	Note
Assumptions		
1	Team members have the necessary skills of teamwork and programming skills	Resources
2	during the covid-19 pandemic convenient for app development	Market
3	All equipment is in good condition	Device
4	There is a mentor to guide	Human
Constraints		
1	Team members must understand and know how to use Java and Python programming languages to develop applications	Technical
2	Team members must understand and know how to use SQLite programming languages to develop database of applications	Technical
3	Team members must understand of AI technology to be able to measure heart rate with the camera	Technical
4	Have full equipment such as: laptop, Macbook, ... for application development	Technical
5	People resource: 4 people	Human
6	The product is aimed at people who have a need for medical examination but are limited in terms of travel or time	Market
7	Total Cost Estimate 4560\$; the application will collect an intermediary for 2% of each transaction made on the system	Cost and benefit

1.4. Project Objectives

1.4.1. Standard Objectives

Metrics	Unit	Committed	Note
Start Date	dd-mm-yyyy	23-08-2021	
End Date	dd-mm-yyyy	14-12-2021	
Duration	days	114	
Team Size	Person	4	
Billable Effort /	Person-day	19	
Number of work hours per day for one engineer	Person-hour	4	

1.4.2. Specific Objectives

Functional goals:

- Works correctly
- Don't have a conflict
- Correct to the requirement

Strategic goals:

- Satisfactory requirement documents
- Website will be available at Viet Nam only

Business goals:

- Income is based on effort
- Website is expected to be released in Dec 2021
- Collect 2% of the medical examination and treatment fee on each medical examination.

1.5. Critical Dependencies (NA)**1.6. Project Risk**

Risk	Description	Probability	Impact	Mitigation Strategy
Lack of coding experiences	Only one member of the team learned Android programming	H	M	Each team member has to learn and help the other to learn quickly.
Source Code conflict	Problems while merging code between members to master branch	H	H	Each team member must resolve conflicts by using git merge CLI before merging to master branch.
Member conflict	Team member maybe conflict with each other while discussing	H	M	Team building, playing board games to get everyone together.
Teamwork tools	Teams maybe have trouble finding the suitable tools for teamwork	H	M	Try to find and learn how to use the tools suitable
Time management	Every member has to study at school.	H	H	Overtime

Probability		Severity	
L	Rarely happened.	L	Low damaged
M	Sometime happened	M	Medium damaged
H	Usually happened	H	Serious damaged

2. PROJECT DEVELOPMENT APPROACH

2.1. Technical Process

2.1.1. Reasons for selecting

Why do we choose the Agile methodology?

Teams pick Agile to adapt rapidly to market changes or client input without losing a year's value. "Enough" planning and delivering in modest, regular increments enables your team to get input on each modification at a minimum expense.

But it's not only a game of numbers - it's about people first and foremost. Authentic human connections are more essential than strict processes, as articulated in the Agile Manifesto. Instead of established arrangements, working with clients and coworkers is more vital. More essential than hyper-detailed documentation is to provide a workable solution to the customer's problem.

An agile team unifies itself around the same vision and then delivers the finest method to life. Each team establishes its own quality, usability and integrity criteria. Their "definition of finished work" indicates how quickly they churn out. Even if it can be intimidating at first, companies' leaders find it more appropriate for the team to achieve or surpass management objectives if they place their faith in a more agile workforce. This team feels more ownership.

The Agile methodology is also straightforward. The rules, objects, actions, and roles are all simple to comprehend. Its approach really aids in the removal of uncertainties in the development process while allowing firms to add their own unique flavor.

It includes frequent releases, usually every four weeks, and the client obtains a fully functional application with more and more functionality added each time. Agile-Scrum supports adaptable, concurrent workflows. It divides projects into smaller, iterative phases, and always makes sure small parts of the project are completed correctly with less risk. Upgrades and edits or early error fixing can therefore be more easily integrated than when using a V-cycle or Waterfall model.

2.1.2. Agile Methodology

Agile methodologies are approaches to product development that are aligned with the values and principles described in the Agile Manifesto for software development. Agile methodologies aim to deliver the right product, with incremental and frequent delivery of small chunks of functionality, through small cross-functional self-organizing teams, enabling frequent customer feedback and course correction as needed.

In doing so, Agile aims to right the challenges faced by the traditional “waterfall” approaches of delivering large products over long periods of time, during which customer requirements frequently changed, resulting in the wrong products being delivered.

Agile methodology is an approach to project management that uses four values to organize projects used as four manifestos:

1. **Individuals and interactions over processes and tools** - The first value emphasizes teamwork and communication. We must understand that software development is a human activity and that the quality of interaction between people is vital. Tools are an important part of software development, but making great software depends much more on teamwork, regardless of the tools the team may use.
2. **Working software over comprehensive documentation** - Documentation has its place and can be a great resource or reference for users and coworkers alike. The main goal of software development, however, is to develop software that offers business benefits rather than extensive documentation.
3. **Customer collaboration over contract negotiation** - Development teams must work closely and communicate with their customers frequently. By listening to and getting feedback, teams will understand what all stakeholders really want.
4. **Responding to change over following a plan** - Changes are a reality in Software development, a reality that your Software process should reflect. A project plan must be flexible enough to change, as the situation demands.

The Agile Process Flow

The Agile Software Life Cycle is the organized set of steps through which a product progresses from start to finish. There are six phases in it:

1. Concept - Projects are envisioned and prioritized
2. Inception - Team members are identified, funding is put in place, and initial environments and requirements are discussed
3. Iteration - The development team works to deliver working software based on iteration requirements and feedback
4. Release - QA (Quality Assurance) testing, internal and external training, documentation development, and final release of the iteration into production
5. Maintenance - Ongoing support of the software
6. Retirement - End-of-life activities, including customer notification and migration

2.1.2.1. Scrum Process

Scrum is one of the working models to apply and implement Agile methodology principles.

Scrum Principles, Phases and Processes

The Scrum method relies on the incremental development of a software application while maintaining a completely transparent list of upgrade or correction demands to be implemented (backlog). It involves frequent deliveries, usually every four weeks, and the client receives a perfectly operational application that includes more and more features every time. This is why the method relies on iterative developments at a constant rhythm of 2-4 weeks. Upgrades can therefore be more easily integrated than when using a V-cycle.

Scrum processes address the specific activities and flow of a Scrum project, which are grouped into the following five phases:

1. Initiate - This phase includes the processes related to initiation of a project: Create Project Vision, Identify Scrum Master and Stakeholder(s), Form Scrum Team,

Develop Epic(s), Create Prioritized Product Backlog, and Conduct Release Planning.

2. Plan and Estimate -This phase consists of processes related to planning and estimating tasks, which include Create User Stories, Approve, Estimate, and Commit User Stories, Create Tasks, Estimate Tasks, and Create Sprint Backlog.
3. Implement - This phase is related to the execution of the tasks and activities to create a project's product. These activities include creating the various deliverables, conducting Daily Standup Meetings, and grooming (i.e., reviewing, fine-tuning, and regularly updating) the Product Backlog at regular intervals.
4. Review and Retrospect - This phase is concerned with reviewing the deliverables and the work that has been done and determining ways to improve the practices and methods used to do project work.
5. Release - This phase emphasizes on delivering the Accepted Deliverables to the customer and identifying, documenting, and internalizing the lessons learned during the project.

Scrum organization

The Scrum framework involves the following three main players:

- Product owner: In most projects, the product owner is the leader of the client's project team. He is the one who will define and prioritize the product features and choose the date and content of each sprint based on values (workloads) that the team communicates to him.
- Scrum Master: He is a genuine facilitator on the project as he makes sure that everyone works at their full potential by eliminating impediments and protecting the team from external interference. Moreover, he pays particular attention to the respect of the different SCRUM phases.
- Team: A team is typically made up of 4-10 people and groups together all the IT specialists who are necessary on a project, i.e. an architect, a designer, a developer,

a tester etc. The team is self-organizing and remains unchanged during an entire sprint.

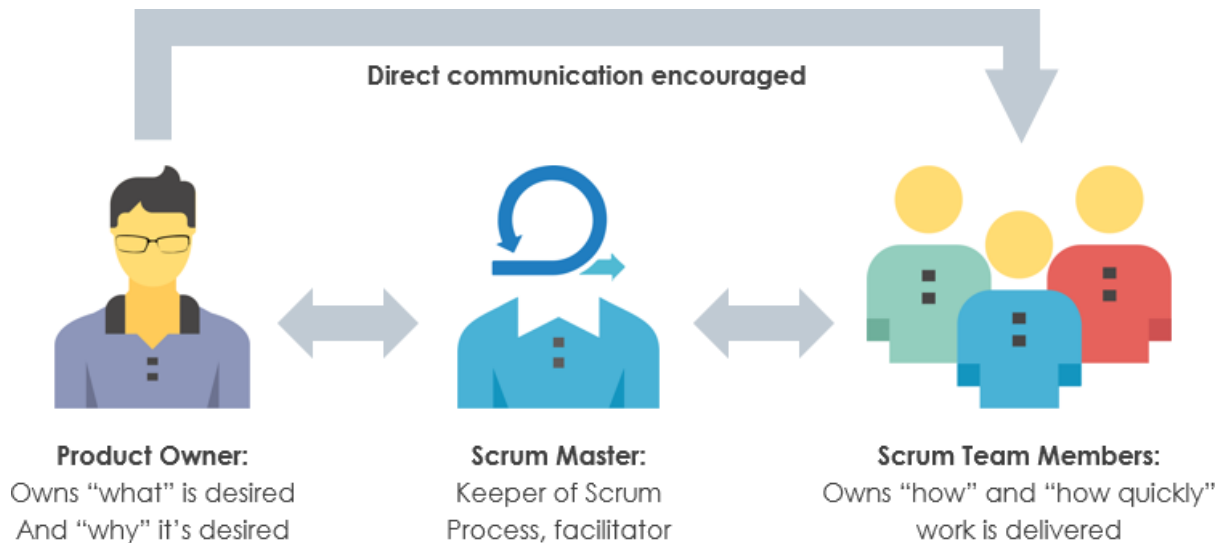


Figure 4.2: Scrum team members

Scrum advantages

In each business there are two significant advantages of using the Scrum technique. The first is that the Scrum teams finish a significantly better quality of their projects. The second greatest advantage of Scrum is that it assures the most effective way for Scrum teams to work.

Scrum efficiently uses time and money resources, since the Scrum teams plan and budget their finances before any project begins. This helps to eliminate unanticipated costs and time delays. Towards the conclusion of each sprint, Scrum teams also test their product for any defects and repair them before release.

A new project has numerous factors and aspects to consider to ensure its success. Scrum techniques make it much simpler to keep a track on these pieces, as large jobs have to be broken up into smaller, more manageable parts and teams.

Projects can evolve and alter over time, which can also affect project needs. Scrum is a change-free strategy since the Scrum teams operate in sprinters, and any modifications required may be quickly incorporated in the following sprint. They must not start at all, which is typically the case with teams who follow the methodology of Waterfall.

Scrum's daily Scrum meetings are another component of the Scrum team. This allows the whole team to monitor the progress of every iteration and to discover any hitches they could face in the project early. This also shows every team member's duties on a daily basis, so that their work is validated.

Because of the positive effects that the Scrum method brings, today this method has become increasingly popular around the world in the software production process.

In short:

Scrum can help teams complete project deliverables quickly and efficiently.

- Scrum can help teams complete project deliverables quickly and efficiently
- Scrum ensures effective use of time and money
- Large projects are divided into easily manageable sprints
- Developments are coded and tested during the sprint review
- Works well for fast-moving development projects
- The team gets clear visibility through scrum meetings
- Scrum, being agile, adopts feedback from customers and stakeholders
- Short sprints enable changes based on feedback a lot more easily
- The individual effort of each team member is visible during daily scrum meetings

Limit and overcome

The Scrum method does not provide a perfect solution to all of the issues that arise during software development. Teams must pay attention to the problem mentioned below, which provide a systematic solution based on extending the approach:

- Team size: team size is typically limited to 4 - 10 people and can become an impediment if it exceeds these recommendations. In the latter case, the organization of meetings becomes impossible and the very foundations of the method are affected. The solution is to set up a Scrum of Scrums. This consists in dividing the project into teams of appropriate sizes and adding an instance of a higher level which groups together the Scrum Master of each Scrum.

- With a whole project of a group of total 4 members, to apply Scrum, the team will divide the roles equally among the members. Each member will be alternately assigned responsibilities, as well as multiple tasks.
- Multiple requests: Requests may be transmitted through several channels on a project and can sometimes be difficult to manage due to their contradictory aspects. These contradictions can slow down the delivery validation process. In order to solve this problem, it is vital to use a single request management tool, which is a standard option with projects.
- Development quality: The more the number of teams increases, the more difficult it becomes to deal with quality. This rule is all the more true when the project is distributed among several centers. The main risks are related to code quality and the number of bugs identified during integration. This is why it is important to have a rigorous quality policy and a project quality plan which precisely defines the rules of the project. Frequent code audits and the implementation of indicators that measure the developers' performance allow minimizing this risk.

2.2. *Quality Management*

2.2.1. *Strategy for Meeting Quality Objectives*

Strategy	Expected Benefits
Do defect prevention using the standard defect prevention guidelines and process; use standards developed in ABC for coding.	10–20% reduction in defect injection rate and about 2% improvement in productivity
Group review of program specs for first few/logically complex use cases. Group review of design docs/first time-generated code by project leader, developer, and one consultant.	Improvement in quality as overall defect removal efficiency will improve; some benefits in productivity as defects will be detected early
Introduction of RUP methodology and implementing the project in iterations. Milestone analysis and defect prevention exercise will be done after each Iteration.	Approximately 5% reduction in defect injection rate and 1% improvement in overall productivity

2.2.2. Quality Control

Review Item	Type of Review	Reviewer	When
Project plan Project	Group review	Mentor	End of Initiation stage
Schedule	Group review		
CM Plan	One-person review		
Product Backlog, User Story	Group review	Mentor	End of 90% of requirements
Design document, object model	Group review	Mentor	End of 90% design
Sprint backlog	One-person review	Scrum Master	Beginning of each stage
Test Case	Group review	Product Owner	End of detailed design
Code	Group review	Mentor	After coding for first few programs

2.2.3. Measurements Program

Data to be collected	Purpose	Responsible	When
Size: No. of KLOC// FP	Evaluate size of the system	SM	At the end of stages
Effort: No. person-day	Evaluate member productivity	Team members	Daily
Quality: No. defects detected	Evaluate, improve the quality of the product	Reviewer, Tester	Right after the review/test
Schedule	Control schedule	SM	Weekly and at the end of stages

2.3. Unit Testing Strategy

- Implemented by the developer.

Completion criteria: Completion criteria are stated to for two purposes:

- Identify acceptance criteria for product quality.
- Identify when the testing is successfully executed

A clear statement of completion criteria should include the following items:

- Function, behavior, or condition being measured
- Method of measurement

Criteria or degree of conformance to measurement Special considerations:

This section should identify any influences or dependencies, which may impact or influence the test effort described in the test strategy. Influences might include:

Human resources (such as availability or need for non-test resources to support / participate in test) Constraints, (such as equipment limitations or availability, or the need / lack of special equipment) Special requirements, such as test scheduling or access to systems

Testing may be stopped when

- It becomes unproductive
- It requires a certain coverage
- It requires a certain number of errors to be found
- Schedule time runs out

2.4. Integration Testing Strategy

N/A

2.5. *System Testing Strategy*

UI Testing Strategy

UI Testing involves testing all visual indicators and graphical icons, including menus, radio buttons, text boxes, checkboxes, toolbars, colors, fonts, and more....

UI testing is performed manually or with an automated testing tool. Regardless of the method used, the goal is to ensure all UI elements meet the requested specifications.

Approaches to UI Testing:

Using manual testing, a human tester performs a set of operations to check whether the application is functioning correctly and that the graphical elements conform to the documented requirements. Manual-based testing has notable downsides in that it can be time-consuming, and the test coverage is extremely low. Additionally, the quality of testing in this approach depends on the knowledge and capabilities of the testing team.

Here is a checklist that UI tests tend to verify:

- Data type errors – Ensure only valid data can be entered for specific data types such as currency and dates.
- Field widths – If a certain text box permits a specified amount of characters, then make it clear on the user interface that the data entered shouldn't exceed the character limit. (For instance, a field that allows 50 characters in the application's database should not allow users to enter more than 50 characters on the interface).
- Navigational elements – Verify all navigational buttons on the page are working correctly, and that they redirect users to the right page or screen.
- Progress bars – When displaying screens that take time to render results, a progress bar should be used to show the user that a process is still running.
- Type-ahead – If your UI uses drop-down lists, ensure you include type ahead. In a drop-down menu with hundreds of items, typing the first letter should skip the list to items beginning with that letter such that users will not have to check through a long list.

- Table scrolling – If data in your tables extends to another page, then the scroll function should allow users to scroll the data but keep all headers intact.
- Error logging – When the system experiences a fatal error, ensure the application writes the error details to an event viewer or log file for later review.
- Menu items – Ensure the application only displays valid menu items that are available at a particular state.
- Working shortcuts – For applications that support shortcuts, verify whether they work correctly, no matter the browser, platform, or device being used.
- Confirm action buttons – Ensure the UI has working confirm button every time the user wants to save or delete an item

In addition to these tests, it is recommended to run UI tests on the significant end-to-end user processes.

➤ UI Testing is completed when the criteria expected in the above checklist are fulfilled.

Functional Testing Strategy

Functional testing is a process of verifying that a system performs as expected when its features are exercised by another system or directly by a user. This means that it lends itself nicely to test case and usage case definitions that can provide a stable, repeatable basis for evaluating the progress of system development.

The entire range of the development process comes under the purview of functionality verification.

Approaches to functional testing:

Using black box testing. It takes an array of inputs and looks for the generation of specified outputs. The idea behind the name is that the contents of the code under test is unknown to the test case and, by definition, to the tester who is only concerned with function verification.

Functional testing techniques:

Functional testing can be broadly divided into two categories – positive and negative testing. Let's take a closer look at each of these categories:

Positive Testing

This type of testing ensures that a program meets the basic requirements of the end-users and runs efficiently upon valid inputs and user flows.

Negative Testing

This type of testing verifies that a program can handle invalid inputs or unintended flows. For example, the software should not crash when entering incorrect characters into a text field.

Functional testing process:

Functional testing will proceed in the following 6 steps:

1. Identify and clarify the functions that you expect the software or web application to perform
 2. Create input data based on these functional specifications
 3. Determine the output based on these functional specifications
 4. Write and execute test cases to gather test data (these can be manual or automated tests)
 5. Compare the results of actual and expected output
 6. Make changes *if* the results don't match the end user's needs
- The goal of this testing is to check whether the system is functionally perfect!

3. ESTIMATION

3.1. Size

Modules	LOC
Frontend	1300
Backend	300
Total	1600

3.2. Effort

	Effort	Position
UI/UX Design	4	Designer
UI Implementation	2	Front-end Developer
BA	1	Business Analyst
PM	1	Project Manage
Development	4	Software Engineer
QA	4	Tester

3.3. Schedule

3.3.1. Project Milestone & Deliverables

No	Phase	Iteration	Start Day	End Day
1	Development	SPR1	Sep 20, 2021	Oct 10, 2021
		SPR2	Oct 11, 2021	Oct 31, 2021
		SPR3	Nov 01, 2021	Nov 21, 2021
		SPR4	Nov 22, 2021	Dec 12, 2020

3.3.2. Detailed Schedule

No.	Task	Task owner	Start	End	Days
1	Start-Up	All member	23/8/2021	19/9/2021	28
1.1	Gathering Requirement	All member	23/8/2021	4/8/2021	2
1.2	Create Proposal Document	All member	25/8/2021	28/8/2021	4
1.3	Project's Kick-off Meeting	All member	29/8/2021	29/8/2021	1
1.4	Create Documents: • Project Plan • Database Design • Product Backlog	All member	30/8/2021	18/9/2021	20
1.5	Review Documents	All member	19/9/2021	19/9/2021	1

2	Development	All member	20/09/2021	12/12/2021	84
2.1	Sprint 1		20/09/2021	10/10/2021	21
1	Sprint Backlog, US, Test Plan, Test Case for Sprint 1	All member	20/09/2021	22/09/2021	3
2	Design UI, Database	All member	22/09/2021	23/09/2021	2
3	Set up Develop Environment	All member	24/09/2021	25/09/2021	2
4	XML & JAVA	All member	26/09/2021	04/10/2021	9
5	Testing	All member	04/10/2021	06/10/2021	3
6	Fix bugs	All member	07/10/2021	10/10/2021	4
2.2	Sprint 2		11/10/2021	31/10/2021	21
1	Sprint Backlog, US, Test Plan, Test Case for Sprint 2	All member	12/10/2021	14/10/2021	3
2	Design UI, Architecture	All member	15/10/2021	17/10/2021	3
3	Update Database	All member	18/10/2021	18/10/2021	1

4	XML & JAVA	All member	19/10/2021	26/10/2021	8
5	Testing	All member	27/10/2021	28/10/2021	2
6	Fix bugs	All member	29/10/2021	31/10/2021	3
2.3	Sprint 3		01/11/2021	21/11/2021	21
1	Sprint Backlog, US, Test Plan, Test Case for Sprint 3	All member	01/11/2021	04/11/2021	4
2	Design UI, Architecture	All member	05/11/2021	07/11/2021	3
3	Update Database	All member	08/11/2021	08/11/2021	1
4	XML & JAVA	All member	09/11/2021	17/11/2021	9
5	Testing	All member	18/11/2021	19/11/2021	2
6	Fix bugs	All member	20/11/2021	21/11/2021	2
2.4	Sprint 4		22/11/2021	12/12/2021	21
1	Complete documents	All member	22/11	25/11/2021	4

2	Design UI	All member	26/11/2021	26/11/2021	1
3	Update Database	All member	27/11/2021	27/11/2021	1
4	XML & JAVA	All member	28/11/2021	07/12/202	10
5	Testing	All member	08/12/2021	09/12/2021	2
6	Fix Bugs	All member	10/12/2021	11/12/2021	2
7	Review Meeting	All member	12/12/2021	12/12/2021	1
3	Project Meeting	All member	13/12/2021	13/12/2021	1
4	Final Release	All member	14/12/2021	14/12/2021	1

3.4. Resource

<i>Position</i>	<i>Member</i>	<i>Effort</i>
<i>Designer</i>	<i>All member</i>	<i>4</i>
<i>Front-end Developer</i>	<i>All member</i>	<i>4</i>

<i>Back-end Developer</i>	<i>Ngô Văn Kha</i> <i>Lê Văn Xuân</i> <i>Trương Gia Huy</i>	3
<i>Data Engineer</i>	<i>Trương Gia Huy</i> <i>Ngo Van Kha</i> <i>Le Van Xuan</i>	3
<i>Tester</i>	<i>All member</i>	4

3.5. Infrastructure

Work/Product	Purpose	Expected Availability by	Note
Development Environment			
NT Server	Operating System	Initiation stage	
Mobile: Android	Operating System	Construction Stage	
Firebase	Real-time Database	Construction Stage	
Java	Development language for App interface	Construction Stage	
Python	Development language for App Backend	Construction Stage	
Hardware & Software			

50GB space on server			
Other Tools			
Google Sheet	Track member activities. At the end of each day, team members will post on time log and scrum master will check.	Construction stage	
Slack	Store document resources and designed components	Definition stage	
Github	Repositories for source code version management.	Definition stage	
Discord	Discuss online, stream and share problems	Definition stage	
Trello	Division of work	Definition stage	

3.6. Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
Technical			
Java Language	All members	14 days	If already trained
Firebase	All members	14 days	If already trained
Process			
Quality system	All member	3 hrs	If already trained
Configuration management	All member	2 hrs	If already trained for
Google Sheet tool	All member	4.5 hrs	Mandatory
Github	All member	5 hrs	Mandatory
Scrum	All member	2 hrs	Mandatory

3.7. Finance

Sprint	Duration(hours)	Salary Rate (USD / Hour)	Cost (\$)/ person
Start-up	140	2.0	\$280
1	105	2.0	\$210
2	105	2.0	\$210
3	105	2.0	\$210
4	105	2.0	\$210
Total	560	2.0	\$1120

4. PROJECT ORGANIZATION

The project team applies the scrum process to project management and development.

4.1. Organization Structure

Full name	Email	Phone number	Role
Ngo Van Kha	ngokha437@gmail.com	0935950384	Scrum Master
Trương Gia Huy	truonggiahuy7799@gmail.com	0972173687	Team Member
Tran Nguyen Thanh Hong	thanhhong24102000@gmail.com	0782363961	Team member

Le Van Xuan	levanxuan5518@gmail.com	0905596460	Team member
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4.2. Project Team

Role	Responsibility	Full name
Product Owner	Understand the user and customers with their needs. Collaborate with the development team. Manage the stakeholders. Describe the user experience and product features. Provides detailed user stories.	Msc Nguyen Tan Thuan
Scrum Master	Communicate the value of Scrum Teach the organization on Scrum to maximize business value Facilitate Sprint Planning, Daily Scrums, Sprint Reviews and Retrospective Meetings Create the Task Board and Sprint Burndown Chart at the start of every Sprint Attend all Scrum meetings Preserve the integrity and spirit of the Scrum framework Maintain the focus of the Team Make the Team aware of impediments and facilitate efforts to resolve them Serve as a coach and mentor to members of the Team	Ngo Van Kha

	Respectfully hold the Team, Product Owner and Stakeholders accountable for their commitments Continually work with the Team and business to find and implement improvements	
Secretary	Record the content of group meetings and activities of the member	Tran Nguyen Thanh Hong
Reviewer	Analysis of the functions and requirements of the product. Review documents related to the project	Truong Gia Huy
Team member	Design, XML Dev, Java Dev, Tester	Truong Gia Huy
Team member	Design, XML Dev, Java Dev, Tester	Le Van Xuan
Team member	Design, XML Dev, Java Dev, Tester	Ngo Van Kha
Team member	Design, XML Dev, Tester	Tran Nguyen Thanh Hong

5. COMMUNICATION & REPORTING

5.1. Communication methodology

Type of Communication	Communication Schedule	Communication way	Who Initiates	Recipient
Status Report (Daily meeting)	Daily	Slack, Discord	Scrum Master	Scrum Team
Schedule and Effort Tracking	Daily	Face to face or GSheets	Scrum Master	Scrum Team
Work Review	Daily	Face to face	Scrum Master	Scrum Team
Work Report	Every Wednesday, Sunday	Face to face or Discord	Scrum Master	Scrum Team
Project Review, ask problems	Every Wednesday, Sunday (flexible)	Face to face, Discord or Zoom	Scrum Master	Scrum Team, Mentor
Ask & Review problems	Anytime	Face to face, Slack, Discord, Zoom	Scrum's Member	Mentor and Scrum Team

5.2. Reporting methodology

Audience/ Attendees	Topic/ Deliverable	Frequency	Method
- Product Owner - Scrum Master - Team Members	Project Progress Review	Weekly	Email, Skype or Zoom
- Product Owner - Scrum Master - Team Members	Explicit Requirement	When needed	Email, Skype or Zoom
- Mentor - Scrum master - Team members	Milestone review	End of each Milestone	Skype, Zoom

- Scrum master - Team members	Daily tasks	Each day	Trello, Messenger, Github
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6. CONFIGURATION MANAGEMENT

No	Tool	Content
1	Google Document	Track the changing of documents & manage versions of documents.
2	Github	Repositories for source code version management.
3	Slack	Store document resources and designed components
4	Trello	Division of work

7. SECURITY ASPECTS

- Project team members and partners agree not to provide or share user information to third parties in any way.
- Research on network attack prevention solutions to ensure data security, avoid being exploited and stolen data by hackers.

REFERENCES

No	Reference item	Issued Date	Source	Note
		dd-mmm-yy		

DEFINITIONS AND ACRONYMS

Acronym	Definition	Note
UI	User Interface	
UX	User Experience	
SPR	Sprint	
SM	Scrum Master	
US	User Story	